Project Executable Files

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie: AI-Powered Learning Assistant
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes (README.md)
3	requirements.txt / package.json / dependency file	Yes (requirements.txt)
4	Database schema / seed files (if applicable)	NA (no database)
5	Environment configuration file (.env.example or similar)	Yes (.env.example)
6	Deployed application URL (if hosted)	NA (runs locally)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	No
10	Demo video or walkthrough (if required)	Yes (link in Step 3)

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
/Edugenie-Learning-assistant
├── main.py               - FastAPI app: routes, CORS, error handling
├── qna.py                - Q&A module (Gemini)
├── explanation_module.py - Explanation module (local T5 + Gemini fallback)
├── summary_module.py     - Text summarizer (Gemini)
├── quiz_module.py        - MCQ quiz generator (Gemini)
├── learning_path.py      - Learning path recommendations (Gemini)
├── /templates
│   └── index.html        - Single-page frontend (HTML + JavaScript)
├── /static
│   └── style.css         - Stylesheet
├── requirements.txt      - Python dependencies
├── .env.example          - Environment variable template
├── .gitignore            - Excludes .env, venv, __pycache__
└── README.md             - Setup and run instructions
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not hosted. Runs locally at http://127.0.0.1:8000
Login Credentials (Demo)	Not applicable (no login required)
Platform / Hosting Provider	Local machine (Uvicorn ASGI server)
Repository Link	https://github.com/mraguram1793-cmyk/Edugenie-Learning-assistant
Demo Video Link	https://drive.google.com/file/d/1z4dTeksL0Opie-5HgN_dDmMtfZ450Ctj/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:
Python 3.10 or above (developed on Python 3.13), pip, Git, an internet connection (for the Gemini API and the first-time download of the Hugging Face model, about 3 GB), and a free Gemini API key from https://aistudio.google.com/app/apikey.

Step 1: Clone the repository: git clone https://github.com/mraguram1793-cmyk/Edugenie-Learning-assistant.git
Step 2: Go to the project folder: cd Edugenie-Learning-assistant
Step 3: Create a virtual environment: python -m venv venv
Step 4: Activate it (Windows): venv\Scripts\activate
Step 5: Install dependencies: pip install -r requirements.txt
Step 6: Configure environment: copy .env.example .env, then open .env and set GEMINI_API_KEY to your key
Step 7: Start the application: uvicorn main:app --reload
Step 8: Open the browser at http://127.0.0.1:8000

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	First start-up is slow because the local LaMini-Flan-T5 model (~3 GB with PyTorch) is downloaded and loaded.	One-time download; later runs use the cached model. If it fails to load, Explanation falls back to Gemini.
2	Quiz output depends on Gemini returning valid JSON; occasionally parsing can fail.	An error message is shown; click Generate Quiz again.
3	Requires a valid Gemini API key and internet access; free-tier rate limits may apply.	Add the key to .env; retry after a short wait if the limit is reached.